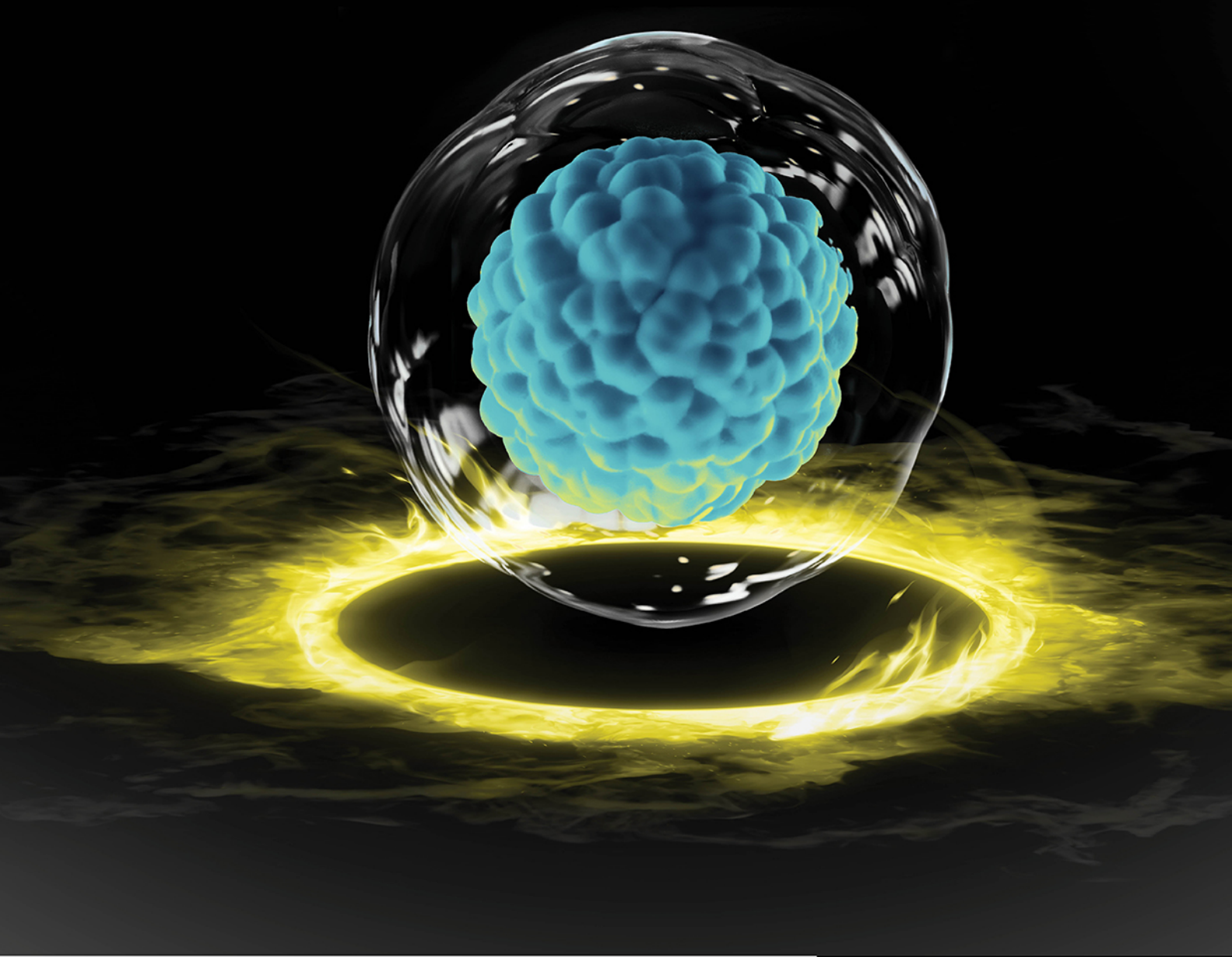




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ARTICLE

Leveraging mathematical models for optimizing filter utility at manufacturing scale

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Abstract

In the production of biopharmaceuticals depth filters followed by sterile filters are often employed to remove residual cell debris present in the feed stream. In the backdrop of a global pandemic, supply chains associated with the production of biopharmaceuticals have been constrained. These constraints have limited the available amount of depth filters for the manufacture of biologics. This has placed manufacturing facilities in a difficult position having to choose between running processes with reduced number of depth filters and risking a failed batch or the prospect of plants going into temporary shutdown until the depth filter resources are replenished. This communication describes a modeling based method that leverages manufacturing scale filtration data to predict the depth filter performance with a reduced number of filters and an increased operational flux. This method can be used to quantify the acceptable level of area reduction before which the filtration process performance is affected. This enables facilities to manage their filter inventory avoiding potential plant shutdowns and reduces the risks of negative depth filter performance.

KEYWORDS

bioprocessing, depth filter, filter fouling modeling, mammalian cell culture

1 | INTRODUCTION

The need for biologics to be produced at an unprecedented pace has meant that biopharmaceutical companies have needed to employ a scalable platform so as to simplify and streamline the development process during development of biological drug candidates (Li et al., 2010). Depth filtration is a unit operation that is ubiquitous across the platform and is a popular option due to the ease of scalability and the availability in a single use format (Shukla & Gottschalk, 2013). Depth filters are typically used in conjunction with a disk stack centrifuge in the upstream portion of the process platform and are predominantly used to remove cells and cellular

debris (Liu et al., 2010). They are also used in downstream unit operations as a turbidity reduction step to remove particulates that may affect the performance of chromatography steps (Brodsky et al., 2012). Depth filters are able to perform these tasks due to their ability to remove particulates and other impurities through their mechanisms of sieving and adsorption which allow for particle removal based on size (Yigzaw et al., 2006) and charge (Singh et al., 2016) respectively. The optimum filter area for manufacturing can be difficult to predict without empirical data. The area can be influenced by a large number of parameters, including mode of operation, type of cell line, level of aggregates, cell culture conditions and centrifuge operating conditions (Zydney & Ho, 2002). During

Abbreviations: A, total filter area; C, volume filtered particles per volume feed filtered; K_c , cake formation coefficient; K_s , standard pore plugging coefficient; L_p , pore length; N_p , number of pores; P, differential pressure; P_a , critical pressure; Q, feed volumetric flow rate; r, pore radius; R_c , resistance due to cake formation; R_m , membrane resistance due to standard pore plugging; s, cake compressibility; t, filtration time; V, volume of feed filtered; μ , viscosity of feed.

process development, bench scale depth filter cartridges are used to mimic the commercial scale depth filter processes performance. These bench scale empirical experiments allow for testing the likely depth filter performance at commercial scale using smaller amounts of feed material (Kim et al., 2021). The bench scale studies are often conducted at constant flux (Goldrick et al., 2017). Optimum filter area required for the commercial scale filter setup is determined by quantifying the cumulative volume of material filtered until endpoints of maximum pressure on the filter (Yavorsky et al., 2003) or maximum turbidity in the filtrate is reached. Furthermore, to account for likely batch to batch variations, the small-scale filtration studies maybe challenged with worst-case centrate generated from the worst case bioreactor conditions. A safety factor of 1.2–1.6× is often applied to the endpoint loading to allow for the filtration process to account for random batch-to-batch process variations (Lutz et al., 2015).

The use of safety factors and the challenge of the filters with the worst-case material allow for a conservative approach of filter sizing and enable a robust manufacturing process with a low chance of pressure increase/yield losses due to filter fouling. However, this conservative approach of oversizing the depth filtration step results in utilizing a redundant amount of depth filters. Oversizing of the depth filtration step can have a negative impact on the filter utilization rate of the constrained depth filter supply chain and as a result can cause bottle necks in the production schedule risking plant shutdowns that could delay production of critical biological material for commercial and clinical purposes. Furthermore, the oversizing of the depth filter also has a negative environment impact by requiring larger than necessary inorganic waste disposal and water required for filter flushes. Empirical filtration studies are often long processes as they match the flux of the commercial process. The duration of these studies limit the testing of filter performance influencing parameters that can be tested during filter sizing experiments. Filter fouling model approaches do not have such a limitation and these approaches allow for insilico testing of filtration process parameters like filter flux. Filter fouling models are typically derived from four classical filter fouling mechanisms: standard, complete, intermediate blocking and cake filtration (Hermia, 1982). In some cases these models are sufficient to describe the fouling processes observed during filtration (Goldrick et al., 2017; Sampath et al., 2014). However, complex biological material consists of an assortment of cellular debris with a range of charges and sizes (Almeida et al., 2022). In cases where the classic filter fouling models do not capture the fouling process, combination models incorporating two or more of the classic filter fouling mechanisms have been shown to describe the fouling behavior of complex biological material (Bolton et al., 2006). This publication uses modeling approaches to understand the fouling behaviour of the depth filter at manufacturing scale by fitting the fouling constants to the available manufacturing empirical data. With these known fouling constants the modeling approaches are then used to predict the reduction of filter area at scale, increasing the availability of depth filters, and therefore the number of batches, when a limited number of filters are available.

2 | MODELING AND THEORETICAL ASPECTS

2.1 | Filter fouling models

The model presented in this report is based on the first principles of Darcy's Law of flow through permeable material. By Darcy's Law, the pressure, flow rate, filter area, resistance, and fluid viscosity are related (Equation 1). Filter fouling by any of the four filter fouling mechanisms (cake, complete pore blocking, intermediate pore blocking, and standard pore plugging) will impact the available filter area or flow resistance through the filter (Hermia, 1982). In work not presented here, we modeled and evaluated the goodness of fit with Akaike's information criterion analysis of the 15 different permeations of the four filter fouling mechanisms to identify the simpler mechanistic model of cell culture centrate fouling over depth filters (Akaike, 1998). We determined that cell culture centrate fouling over depth filtration is best characterized by the combination of a compressible cake formation and standard pore plugging filter fouling. We assume that over the course of the filtration, the particle load and the particle size distribution does not change. We also assume that for the purposes of filter sizing, the tortuous path and heterogeneous pore size distribution of depth filters can be estimated as homogeneously sized cylinders.

$$\frac{P(t)}{Q(t)} = \frac{\mu * R(t)}{A} \quad (1)$$

2.1.1 | Standard pore plugging

The standard pore plugging mechanism applies to particles that are smaller than the size of the pores. Over the course of the harvest, particles enter the pore and become entrapped along the walls of the pores, restricting the pore diameter. The following assumptions are made to simplify the mathematical representation of the particles becoming entrapped within the depth filters. The pores are assumed to be cylindrical with an average effective pore radius r , pore length L_p , and have an average pore dispersion of N_p/A . Also, the volume of particles entrapped in the pores is assumed to be proportional to the volume of material passed through the depth filter. This proportionality constant is referenced as C in Equation 2 below. With these simplifications, we can express the pore radius constriction with a material balance differential equation (Equation 2) and simultaneously use the Poiseuille equation to convert the pore radius to the flow resistance (Equation 3) (Hermia, 1982).

$$N_p * 2\pi r(t) * L_p * \frac{d}{dt}r(t) = -C \frac{d}{dt}V(t) \quad (2)$$

$$Q(t) = N_p \left(\frac{\pi}{8} * \frac{r(t)^4 * P(t)}{\mu * L_p} \right) \quad (3)$$

By solving the material balance and Poiseuille equations with respect to the initial conditions and time t , it is possible to express the ratio of flux to transmembrane pressure as a function of time to the initial ratio of flux to transmembrane pressure and the total volume filtered (Equation 4). This form of the equation allows for differential pressure to vary as a function of time.

$$\sqrt{\left(\frac{Q(t) * P(0)}{P(t) * Q(0)}\right)} = 1 - K_s * V(t) \quad (4)$$

Where:

$$K_s = \frac{C}{N_p * \pi * L_p * r(0)^2} \quad (5)$$

The maximum volume that can be filtered through the filter before the pores are completely blocked is solved for when either volumetric flow rate approaches zero or the differential pressure approaches infinity:

$$V_{\max} = \frac{1}{K_s} \quad (6)$$

2.1.2 | Cake resistance

Cake filtration occurs when the particles are too large to fit into the pores, but do not block the pores. Instead, a layer of compressible permeable material accumulates on top of the depth filter material. The cake has a resistance to flow proportionate to the thickness of the cake. The cake may compress as the pressure differential across the cake increases. The flow resistance through the cake increases with cake compression. The resistance can be expressed below in Equation 7 (Carman, 1934).

$$R_c(t) = \left(1 + \frac{P(t)}{P_a}\right)^s K_c V(t) \quad (7)$$

2.1.3 | Standard pore plugging with compressible cake filtration

The membrane resistance due to standard pore plugging and resistance due to cake formation can be linearly combined (Equation 8) to create an equation that calculates the pressure drop due to both forms of fouling.

$$R(t) = R_m(t) + R_c(t) \quad (8)$$

$$R(t) = \frac{A/\mu}{(1 - K_s * V(t))^2} * \frac{P(0)}{Q(0)} + \left(1 + \frac{P(t)}{P_a}\right)^s K_c V(t) \quad (9)$$

Using Darcy's law to convert resistance to a final expression for pressure drop across the filter:

$$P(t) = \frac{Q(t)}{(1 - K_s * V(t))^2} * \frac{P(0)}{Q(0)} + \left(1 + \frac{P(t)}{P_a}\right)^s K_c V(t) \frac{Q(t)\mu}{A} \quad (10)$$

2.2 | Model fitting and prediction process

Parameter fitting can be performed using data collected at either constant pressure or constant flow rate. The parameter fitting of the standard pore plugging with compressible cake filtration model was performed by using the generalized reduced gradient nonlinear solver of Microsoft Excel. The sum of square errors of the observed differential pressure versus the calculated differential pressure at each volume reading was minimized by varying K_s , K_c , P_a , and initial ratio of differential pressure to flow rate ($P(0)/Q(0)$) all constrained to be non-negative, and s to be bound between 0 and 1. The nonlinear optimizer was primed with linear regression fittings for K_s and K_c using simplified models of fouling occurring by only standard pore plugging or incompressible cake ($s = 0$). In many cases, the instantaneous volumetric flow rate would be calculated from the time differential of the accumulated filtrate volume measured with a scale. There can be oscillations in this calculation due to scale resolution, strain on the connecting tubing, or scale not being level. To minimize the effect that this oscillation could have on the fitting, we integrated Equation 10 with respect to time. In the case the ratio between the cake compressibility and the critical pressure s/P_a is «1 and the ratio between the differential pressure and critical pressure $P(t)/P_a$ is «1, we can simplify the integration of Equation 10. This results in the following equation in terms of the accumulated filtrate volume, standard pore plugging correlation, cake formation correlation, initial differential pressure, initial volumetric flow rate, and the differential pressure time course as shown below in Equation 11.

$$\int P(t)dt = \frac{1}{K_s * (1 - K_s V(t))} * \frac{P(0)}{Q(0)} + \left(1 + \frac{P(t)}{P_a}\right)^s \frac{K_c V(t)^2 \mu}{2 A} \quad (11)$$

Equation 11 can be used instead of Equation 10 for nonlinear fitting provided that the data points are frequent enough for numerical summation to be an accurate approximation of integration.

$$K_{s,new} = K_{s,old} \frac{A_{old}}{A_{new}} \quad (12)$$

K_s , K_c , and the intercept, $P(0)/Q(0)$ can be shown to be dependent on the available filter area. When predicting the impact of changes in filter area on the predicted differential pressure profile, K_s , K_c , and the intercept are adjusted as shown in Equation 12, Equation 13, and Equation 14.

$$K_{c,new} = K_{c,old} \frac{Q_{old}}{Q_{new}} \quad (13)$$

$$\left(\frac{P(0)}{Q(0)}\right)_{new} = \left(\frac{P(0)}{Q(0)}\right)_{old} \frac{A_{old}}{A_{new}} \quad (14)$$

2.2.1 | Performance prediction

Historical volume-differential pressure data from the same cell line can be used to perform nonlinear regression to determine K_s , K_c , P_a , s , and $P(0)/Q(0)$. K_s , K_c , P_a , and s are inherent functions of particle size distribution, so the historical data of choice is important in achieving accurate results. Corresponding differential pressure profiles can be predicted if any changes are made to harvest size, total filter area, or flow rates. Once the changes are implemented, the resulting differential pressure profile can be compared with the prediction to evaluate the predictability of the model.

3 | MATERIALS AND METHODS

3.1 | Cell culture/centrifugation parameters

Cell culture material used for these experiments was generated using mammalian cell lines where each project represents a different monoclonal antibody (mAb) product. The cultures were produced in pilot scale (2000 L) and commercial scale (12000 L) bioreactors which were harvested during Days 11–14, post-stationary growth phase.

The centrates were generated after processing the cell culture material through Alfa-Laval centrifuges which were operated at 7 L/min and 9500 RPM for the 2000 L scale and at 60 L/min and

TABLE 1 Process conditions of harvested material used for filter fouling analysis.

Material-ID	Scale of operation (L)	Centrifugation conditions: bowl speed (RPM)	Centrifugation conditions: flow rate (L/min)	Depth filter flux at platform operation (LMH)	Depth filter flux post-filter reduction (LMH)
Project-A	12,000	5100	60	55	64
Project-B	2000	9500	7	48	54
Project-C	2000	9500	7	27	32
Project-D	2000	9500	7	27	32

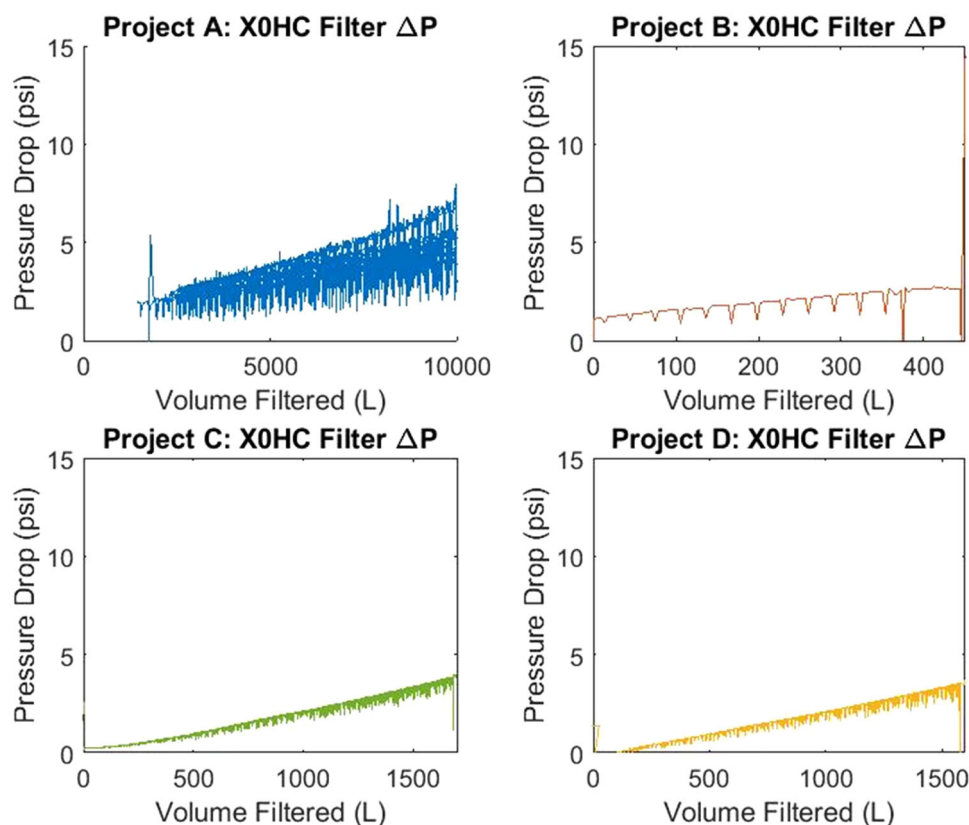


FIGURE 1 Historical data for pressure drop across the depth filter over the course of a harvest (pre-depth filter reduction). Data from multiple historical batches are shown for project A while data from single batches are shown for projects B, C, and D.

5100 RPM for the 12000 L scale. The bioreactor scale and centrifugation process conditions are summarized below in Table 1.

3.2 | Depth filtration setup and conditions

Filtration at pilot and commercial scale was conducted using multiple 1.1 m² Millistack X0HC depth filter capsules set up in series and were flushed with WFI before the loading of centrate. Flow rates were controlled to be constant throughout the entirety of a harvest operation.

4 | RESULTS AND DISCUSSION

4.1 | Modeling approach

The conventional approach of reducing the filters at a commercial facility would involve reducing the 50% safety factor incrementally at scale and subsequently evaluate the performance of the depth filtration process with the reduced area to determine if the process change was either successful or unsuccessful. In the case the depth filter pressures achieved during a live campaign are below maximal allowable pressure and there is no significant increase in subsequent

membrane filter differential pressure then this is deemed a positive performance and further reduction to the safety factor can be enacted. In the case either the depth filter pressure is higher than the maximal allowable pressure or there is a significant increase in subsequent membrane filter differential pressure then the performance is deemed negative and would require reverting back to a setting with a higher filter area and lower flux where successful performance was achieved. This conventional approach to reduce filter area is iterative in nature, making it timely and has inherent risks of failure associated with it. The proposed approach in this communication bridges this risk by leveraging the existing pressure, flux and loading data at manufacturing scale, where the filter size has a 50% safety margin. The pressure flux and loading data from available manufacturing scale runs is used to fit cake and standard pore fouling coefficients in the framework of equations described in Section 2. With the determined coefficients appropriately scaled, the equations are used to generate the pressure profile as a function of loading with a new filter area and new operational flux. The generated pressure and filter loading profiles are evaluated to provide technical guidance on if a change in filter area and/or flux will result in a pressure increase above a given threshold. Filters typically have maximum differential pressure ratings beyond which operation is not possible, however this maximum value is based on ensuring structural integrity of the depth filter and the equipment

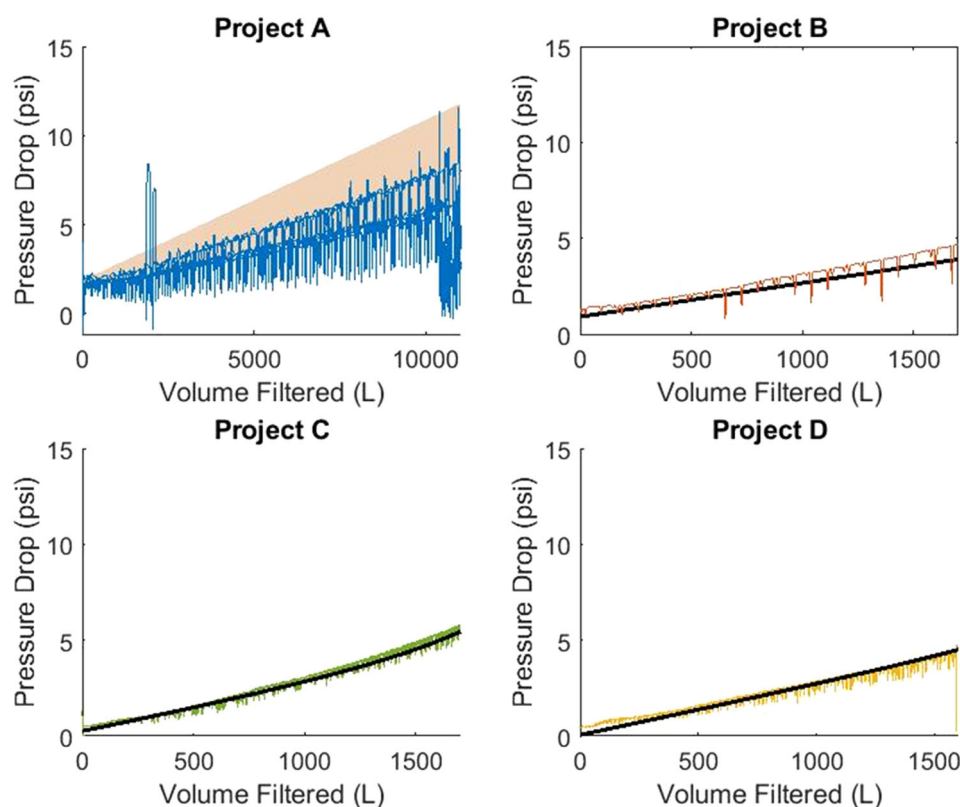


FIGURE 2 A priori model predictions overlayed with post-depth filter reduction harvest data. Project A predictions represented as a colored range due to the availability of multiple historical data sets. Projects B, C, and D a single prediction curve due to availability of just one historical data set.

holding the filters in place. Additionally, the model generated pressure profiles do not take into account for filter fouling occurring via turbidity break through. As a result the studies described in this communication take a conservative value of 10 psi as the pressure threshold.

4.2 | Historical data for filter sizing efforts

Historical data at scale were available for the various projects as shown in Figure 1. The data represent differential pressure across the depth filter over the course of a harvest. The harvest scale depicted in these data ranges from 2000 to 12000 L. All harvests were conducted at platform operation using the standard number of X0HC filters that provided a 150% safety factor based on small scale

studies. For project A, data from multiple batches were available as shown by the multiple pressure profiles. For each project, this historical data were used to build predictive models to predict differential pressure profiles for a reduced number of filters. Empirical parameters K_s , K_c , P_0/Q_0 , s , and P_a were determined using nonlinear regression.

4.3 | Modeling predictions with reduced filter area

X0HC area reductions between 15% and 27% were implemented based on modeling results. Figure 2 shows the a priori model predictions along with the post-reduction harvest data. Model predictions were presented before harvest. This was used to evaluate the predictability of the model. Project A has a prediction range due

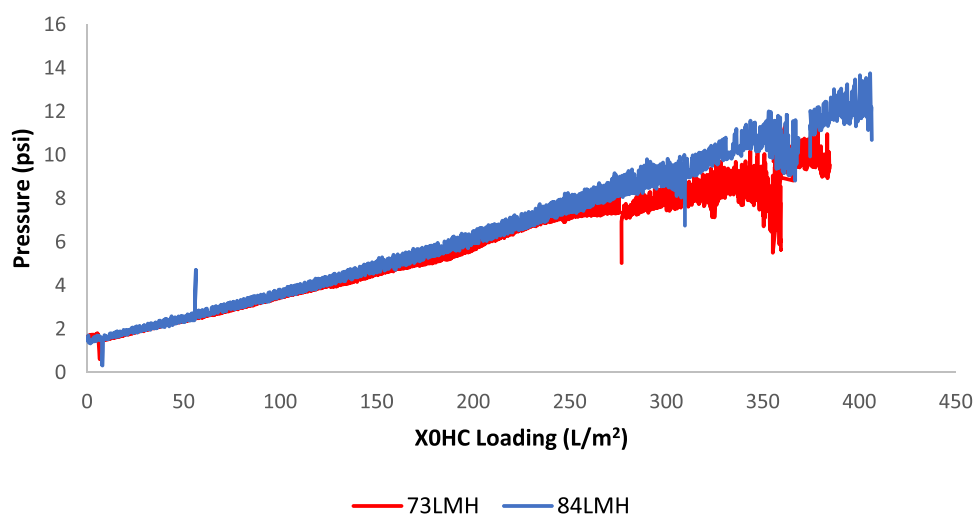


FIGURE 3 Project A small-scale study depth filter pressure drop at fluxes: 73 and 84 LMH.

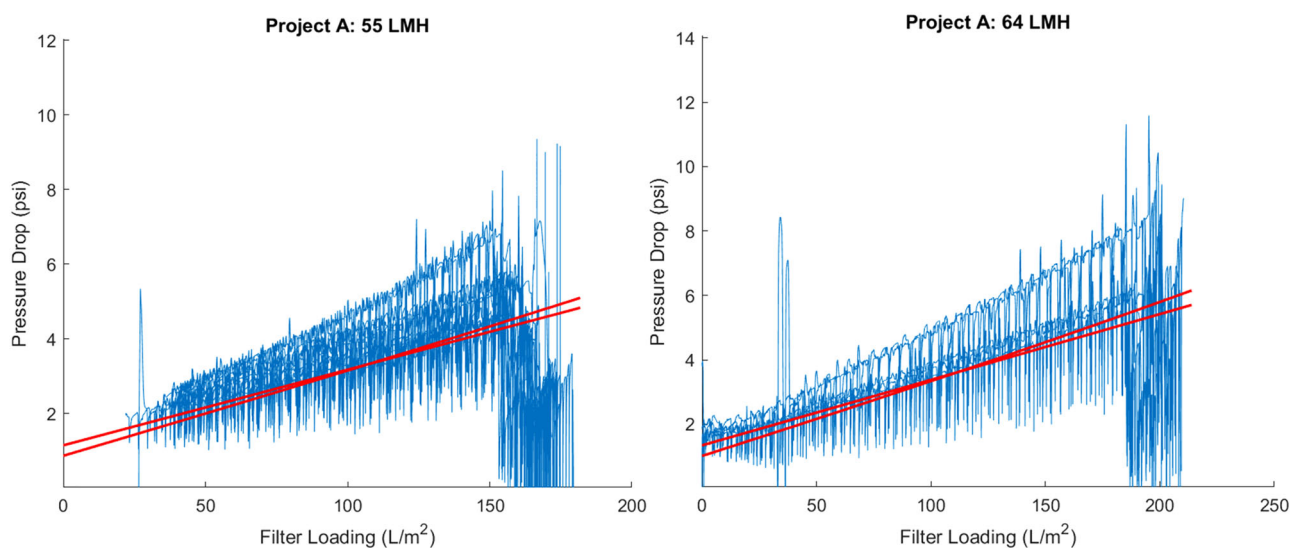


FIGURE 4 A priori model predictions from small-scale data and overlaid with actual large-scale harvest data for Project A at 55 LMH and 64 LMH. The large-scale harvest data are also displayed in Figures 2a and 3a.

to data availability from multiple historical batches. Projects B, C, and D have a single prediction curve since data from only one historical batch was available for analysis. As seen from the prediction versus data comparison in Figure 3, the model is able to very accurately predict pressure drop profiles from historical data.

4.4 | Small scale data and model limitations

A small-scale study for Project A was conducted at two different fluxes: 73 and 84 LMH. The scale of operation for this study was 1 L. To further demonstrate the robustness of the model, both small-scale X0HC pressure profiles in Figure 3 were fitted to the filter fouling model to derive fitting coefficients K_s , K_c , and P_0/Q_0 . The fitting coefficients were then scaled and used to form predictions for commercial scale pressure profiles at 55 and 64 LMH. The commercial scale data for these fluxes are shown in Figure 1 and 2, respectively. The model predictions overlayed on the manufacturing scale data are shown below in Figure 4. These model predictions were made before manufacturing scale harvests. From the data presented in Figure 4, it can be determined small-scale experimental data are in line with commercial scale harvests even at different fluxes. Based on experimental experience at small scale turbidity breakthrough has been observed and identified as a risk. Turbidity breakthrough could result in pressure build up on the sterile filter downstream of the depth filter due to increased flux. One limitation of the model is that it does not predict this phenomena. This risk could be mitigated by monitoring differential pressure on the sterile filter and having back up filters on hand to swap when a pre-determined pressure limit is reached.

5 | CONCLUSION

This report provides an enhanced method for performing at scale filter sizing when filter fouling is occurring by a combination of standard pore plugging and cake formation mechanisms. The equations based on the fouling mechanisms were fitted with the manufacturing and pilot scale filtration data generated with a 50% safety margin. Subsequently these fits were used to predict the likely filtration performance when the operational flux is increased and the available filter area was also simultaneously reduced below the 50% safety margin to a 10%–25% safety margin, depending on the project. The model was shown on multiple occasions to safely reduce the number of filters required in many pilot and commercial scale harvests by successfully predicting expected differential pressure profiles. While the use of this model in this communication focused on the depth filtration during harvest post disk-stack centrifugation, it is also applicable to other filter processes where similar fouling takes place. The utility of the modeling approach described in this communication enables a systematic method of managing the risk of efficiently using the depth filter supply chain resources against

maintaining the performance of the harvest step. The efficient use of the depth filters allows for facilities to avoid costly plant shut downs and enables manufacturers to meet the market demand for biologic therapeutics.

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